

Visit <https://nodejs.org/en/download/> to learn more about how to get it up and running.

Once you are done with the installation process, let us create an empty directory named *ToDoApp* to configure our development environment. Initially, we will start npm, our package manager. To do so, open up your console (in my case, the command prompt) in this directory and run the following command:

```
npm init
```

Fill out the information required to initialise the project (or you can simply skip it by pressing *Enter* until the end), as prompted by the command prompt. For example, see Figure 2.

Now, as we are done with this step, we will have a new file created in our directory named *package.json*, which will contain the project details as shown in Figure 3.

Within the *ToDoApp* directory, create a new file named *index.js* – this is where most of our code will reside. It will have the startup code for our app with routing and other functions of our application.

We will use Express to manage and run the Web server. Express is a flexible and minimal Web framework of Node.js, which makes it really easy to create and run a Web server with Node.

To install Express, we will run the following command:

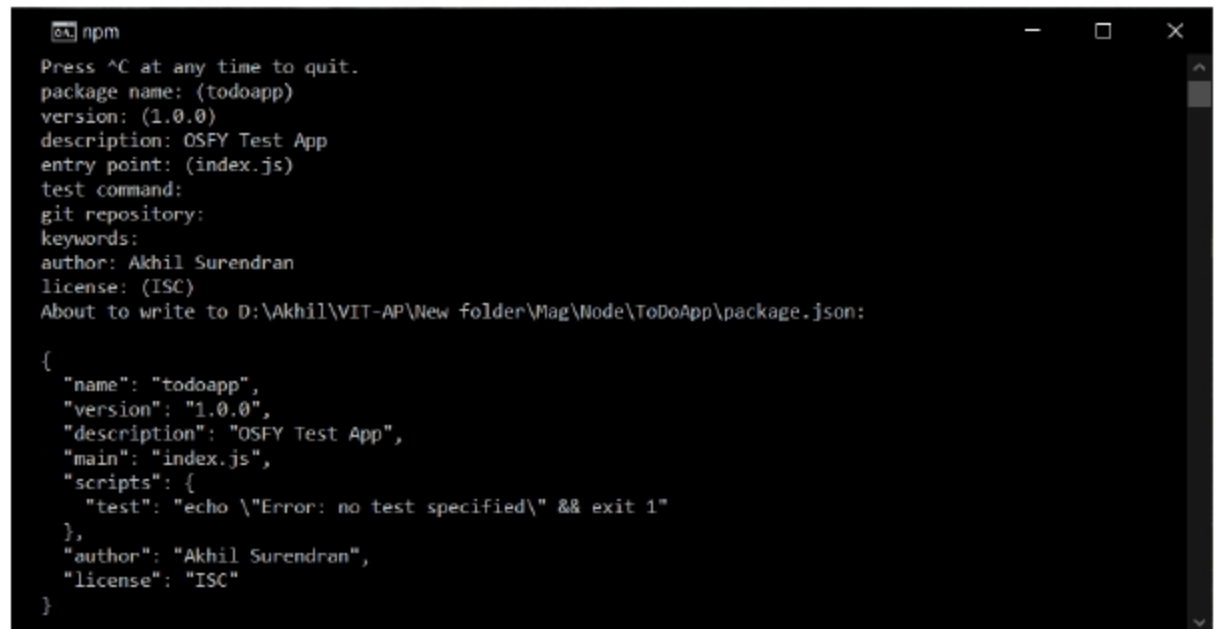
```
npm install express --save
```

After the installation, we can add the following code to the *index.js* file:

```
//require the express app
var express = require('express');

//to call express
var app = express();

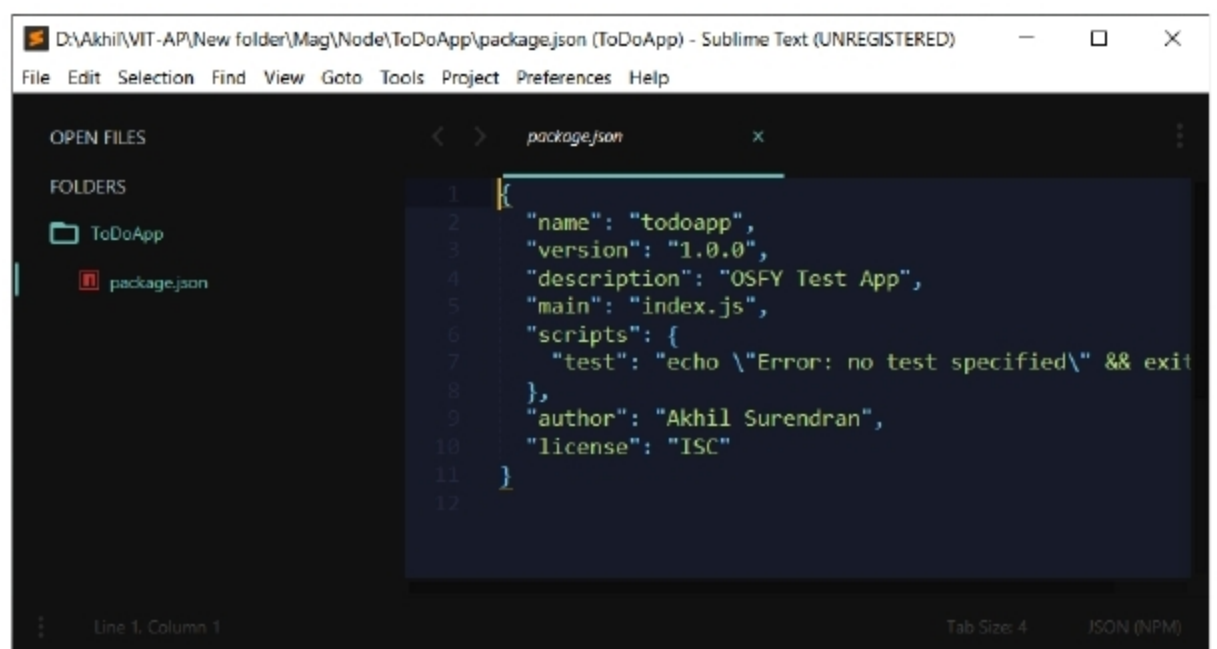
//takes us to the root(/) URL
app.get('/', function (req, res) {
  //when we visit the root URL express
```



```
npm
Press ^C at any time to quit.
package name: (todoapp)
version: (1.0.0)
description: OSFY Test App
entry point: (index.js)
test command:
git repository:
keywords:
author: Akhil Surendran
license: (ISC)
About to write to D:\Akhil\WIT-AP\New folder\Mag\Node\ToDoApp\package.json:

{
  "name": "todoapp",
  "version": "1.0.0",
  "description": "OSFY Test App",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "author": "Akhil Surendran",
  "license": "ISC"
}
```

Figure 2: Initialising the project



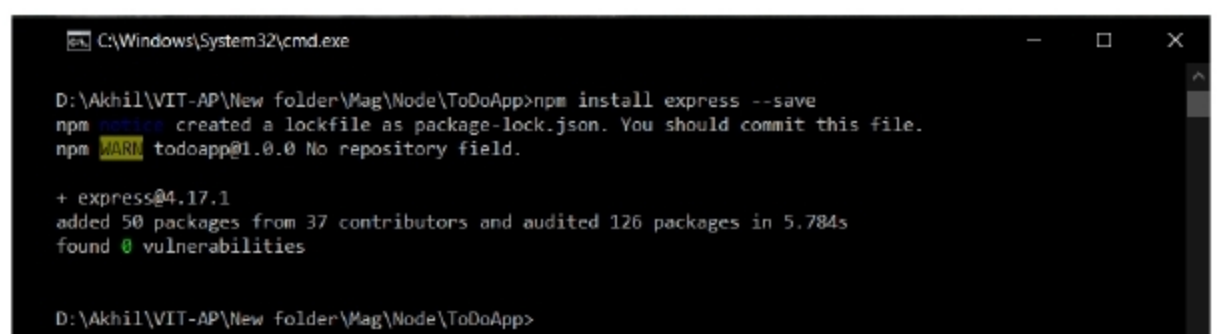
```
D:\Akhil\WIT-AP\New folder\Mag\Node\ToDoApp\package.json (ToDoApp) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
package.json x

FOLDERS
ToDoApp
package.json

1 {
2   "name": "todoapp",
3   "version": "1.0.0",
4   "description": "OSFY Test App",
5   "main": "index.js",
6   "scripts": {
7     "test": "echo \"Error: no test specified\" && exit 1"
8   },
9   "author": "Akhil Surendran",
10  "license": "ISC"
11 }
12
```

Figure 3: *package.json* and our project file directory

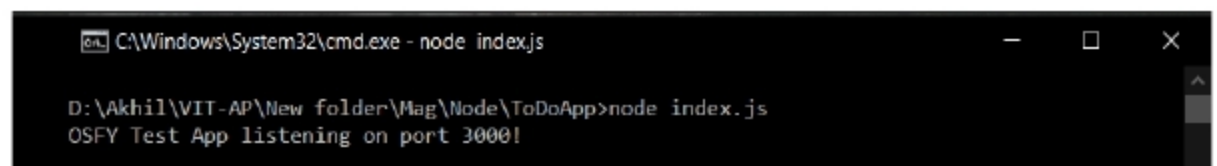


```
C:\Windows\System32\cmd.exe
D:\Akhil\WIT-AP\New folder\Mag\Node\ToDoApp>npm install express --save
npm WARN created a lockfile as package-lock.json. You should commit this file.
npm WARN todoapp@1.0.0 No repository field.

+ express@4.17.1
added 50 packages from 37 contributors and audited 126 packages in 5.784s
found 0 vulnerabilities

D:\Akhil\WIT-AP\New folder\Mag\Node\ToDoApp>
```

Figure 4: Installing a package



```
C:\Windows\System32\cmd.exe - node index.js
D:\Akhil\WIT-AP\New folder\Mag\Node\ToDoApp>node index.js
OSFY Test App listening on port 3000!
```

Figure 5: Testing the server

```
will respond with 'Hello World'
res.send('Hello World!');
});
```

```
//the server is listening on port 3000
for connections
```

```
app.listen(3000, function () {
  console.log('OSFY Test App listening
on port 3000!')
});
```

We will now test our server. To do so, enter the following command in